UNIVERSITY OF WOLLONGONG AUSTRALIA

Tutorial 05

Frequency response and experimental inference

Preparation: Lecture 05: sinusoidal steady state, amplitude and phase, Bode corners and bandwidth; distinguish steady state from startup.
Submission credit: 1% of the course grade for submitted work; correctness is not marked.
In class: Q1–Q3. Extensions are independent practice. Show reasoning, substitutions and units.

Question 1 Quick checks on sinusoidal response and Bode factors

Give a brief reason or calculation.
a) Gain in decibels. A normalized transfer function has DC gain 2. Find its low-frequency gain in dB. What dB change results from multiplying its gain by 10?

b) Sinusoidal steady state. For $G(s)=2/(s+2)$ and input $u(t)=3\sin(2t)$, find the output amplitude and phase after transients decay.

c) Bode corners. In $G(s)=4(1+s/8)/[(1+s)(1+s/40)]$, identify each pole/zero corner and the asymptotic magnitude slope in the four frequency bands.

d) Bandwidth and tracking. For $G(s)=6/(1+0.25s)$, find the −3 dB bandwidth and the largest frequency for which the output amplitude is at least 90% of its low-frequency value, keeping the input amplitude fixed.

Question 2 A fan whose unbalance force grows with speed

A fan assembly is modelled by $Mx'' + cx' + kx = F$, with M = 8 kg, c = 32 N s/m and k = 3200 N/m. Its rotating unbalance has mass–eccentricity product me = 0.004 kg m. At angular speed ω, use the supplied force law $F(t) = me\,\omega^2 \sin(\omega t)$. Ignore startup transients.

a) Find the force amplitude and complex force-to-displacement response at ω = 10 and 40 rad/s.

b) Calculate displacement amplitude and phase lag at both speeds. At 40 rad/s, the compliance |X/F| is smaller than at 10 rad/s, yet the vibration amplitude is larger. Explain why.

c) Find the natural frequency and damping ratio.
MATLAB verification. Calculate force amplitude, displacement amplitude in mm and phase in degrees at 10 and 40 rad/s. Submit one figure with two panels, each showing two periods of its steady displacement response. Label speed, time and displacement. Base MATLAB is sufficient.

Question 3 Infer a zero from a Bode measurement

A normalized measured Bode response is shown. Its low-frequency gain is approximately 9.54 dB. Two candidate models are given below. An independent reading at 10 rad/s is magnitude 1.20 ± 0.05 and phase −42° ± 2°.

$$G_A = \frac{3}{\left(1+\frac{s}{2}\right)\left(1+\frac{s}{20}\right)},\quad G_B = \frac{3\left(1+\frac{s}{5}\right)}{\left(1+\frac{s}{2}\right)\left(1+\frac{s}{20}\right)}$$

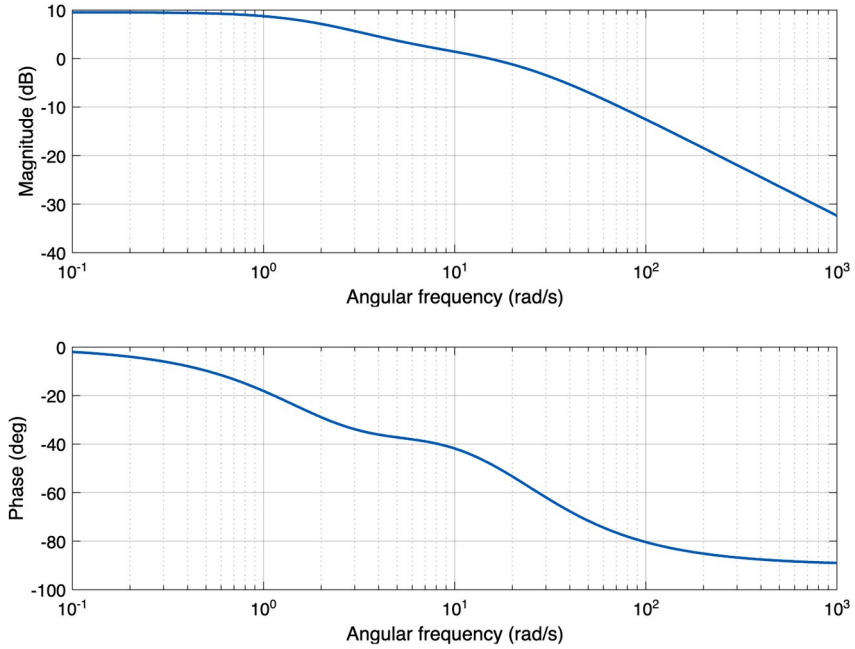


Figure 1. Measured Bode response for comparison with the two candidate models.

a) Read the corner frequencies and slope changes from the plot. Select the matching candidate model and justify your choice.

b) The selected model predicts magnitude 1.1767 and phase −41.820° at 10 rad/s. Compare these supplied predictions with the independent reading. State one limitation of fitting a model from a finite-frequency record.

Extension A: Reconstruct and verify a second-order model from Bode data

After class / optional extension

A force-to-displacement frequency sweep of a mass–spring–damper plant gives a low-frequency compliance of 0.020 m/N. At 10 rad/s the phase is −90° and the compliance magnitude is 0.040 m/N. An independent reading at 20 rad/s is 0.0063 ± 0.0003 m/N with phase −162° ± 3°. Assume a positive-parameter model with force input and displacement output.

a) Use the low-frequency magnitude, the −90° phase crossing and the magnitude at that crossing to infer k, natural frequency and damping ratio. Explain what each reading identifies.

b) Find m and b, then write the force-to-displacement transfer function with units.

c) Predict magnitude and phase at 20 rad/s and compare with the independent reading. Name one reason agreement at these frequencies does not prove the model is exact everywhere.

Method reference

Quick-check sources: Reference/tutorials/Tute 7 Tutorial 7 Questions, Q1–Q2 (dB gain and sinusoidal response). Ogata, Modern Control Engineering, 5th ed., Sec. 7–2, pp. 403–409. Q1 uses independently adapted values; the previous bandwidth task is retained as Q1(d).

Method source: Ogata, System Dynamics, 4th ed., Section 9-3, pp. 440–441 (rotating unbalance). Problem A-9-5, pp. 463–464 illustrates frequency-response parameter identification for a first-order spring–damper model; the second-order extension here adapts that method using Lecture 05. Questions use independently adapted data. Page numbers refer to printed pages.

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